

GETRA

Geo-Enabled Transit & Retail Analytics

BACKEND SYSTEM DOCUMENTATION API REFERENCE & TESTING MANUAL

Phase 16

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-

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1. Executive Summary

The GETRA backend is a robust spatial analytics platform designed for local transit and economic accessibility (UMKM/POI). It features a robust PostgreSQL/PostGIS foundation powered by Supabase, Next.js API routes, and a complete authentication mechanism via Supabase Auth. The foundation for data ingestion, network routing (pgRouting), and spatial boundaries is fully implemented but currently operates on DUMMY test data.

2. System Overview

Component Status Summary

COMPONENT	STATUS	TEST STATUS	NOTES
Authentication	IMPLEMENTED	PASS	Rate limits identified upstream.
Profile	IMPLEMENTED	PASS	User profiles sync via auth triggers.

COMPONENT	STATUS	TEST STATUS	NOTES
Roles	IMPLEMENTED	PASS	RBAC (ADMIN, COMMUTER, UMKM, COMMUNITY).
RLS	IMPLEMENTED	PASS	Row-level security restricts dummy data access.
API Foundation	IMPLEMENTED	PASS	Health, spatial, auth, ingestion APIs ready.
PostGIS	IMPLEMENTED	PASS	BBox and distance queries verified.
pgRouting	IMPLEMENTED	PASS	Node mapping present; production routes blocked.
Spatial Engine	IMPLEMENTED	PASS	Nearby search functionality intact.
Data Ingestion	FOUNDATION_ONLY	PASS	Admin protected ingestion pipeline setup.
Study Area	IMPLEMENTED	PASS	Public API restricted by RLS on dummy data.
Transport	IMPLEMENTED	PASS	Corridors & Nodes (RLS protected).
Pedestrian Network	IMPLEMENTED	PASS	Network tables exist (DUMMY).
UMKM	IMPLEMENTED	PASS	Internal nearby API works.
POI	IMPLEMENTED	PASS	Internal nearby API works.
Survey & Demand	FOUNDATION_ONLY	N/A	Tables created, API pending.
Golden Dataset	IMPLEMENTED	PASS	v1 Dummy established.

COMPONENT	STATUS	TEST STATUS	NOTES
Docker	IMPLEMENTED	N/A	Docker setup is present for isolated run.
Security	IMPLEMENTED	PASS	Protected against direct service role leak.

3. Technology Stack

- **Runtime:** Node.js (>=22.13.0)
- **Framework:** Next.js (16.3.1)
- **Language:** TypeScript (5.9.3)
- **Validation:** Zod
- **Database:** PostgreSQL (via Supabase)
- **Auth:** Supabase Auth
- **Spatial:** PostGIS
- **Routing:** pgRouting
- **Testing:** Vitest
- **Container:** Docker Compose

4. Backend Architecture

The application uses Next.js Route Handlers as the primary API mechanism: Route Handler -> Schema Validation (Zod) -> Repository Layer -> Supabase Client -> PostgreSQL (PostGIS)

5. Project Structure

```

app/api/      # Next.js Route Handlers
src/          # Business logic, repositories, lib functions
supabase/     # Database migrations, seed data, and schema definitions
scripts/      # Provisioning, CLI tasks, regression testing scripts

```

```
tests/      # Unit and integration tests (Vitest)
docs/       # Documentation and phase evidences
```

6. Environment Configuration

VARIABLE	REQUIRED	DESCRIPTION	SECRET?
NEXT_PUBLIC_SUPABASE_URL	YES	Supabase instance URL	NO
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	YES	Supabase anon/publishable key	NO
SUPABASE_SERVICE_ROLE_KEY	NO	Admin key for backend tasks	YES
GETRA_TEST_USER_PASSWORD	NO	Password for dummy tests	NO

7. Running Backend

Development Mode

```
npm run dev
```

Development Base URL: `http://localhost:3000` *Note:* This runs the Next.js dev server. It is not a 24/7 production server.

Production Build

```
npm run build
npm run start
```

To run 24/7, the backend requires a host server, VM, or cloud container platform.

8. Authentication

Client -> POST /api/auth/login -> Supabase Auth -> JWT Session (Bearer Token/Cookie)

9. User Profiles & Roles

Roles implemented: `COMMUTER` , `UMKM` , `COMMUNITY` , `ADMIN` . Profiles are automatically created on registration via a Postgres trigger.

FEATURE / API	PUBLIC	COMMUTER	UMKM	COMMUNITY	ADMIN
/api/health	YES	YES	YES	YES	YES
/api/auth/me	NO	YES	YES	YES	YES
/api/admin/*	NO	NO	NO	NO	YES

10. RLS & Security

Row-Level Security (RLS) protects data isolation. Service roles bypass RLS. *Service role is server-only. Never expose as `NEXT_PUBLIC` or in the client bundle.* Currently, the public dummy data requires authenticated RLS access to read `study_areas` and `transport` tables.

11. API Architecture

Base URL is `http://localhost:3000/api` . All data exchanges use `application/json` .

12. Complete API Catalog

API Index

METHOD	PATH	FULL URL	AUTH	TEST CASE	EXPECTED	ACTUAL	PASS/FAIL
GET	/api/health	http://localhost:3000/api/health	N	System health check	200		
GET	/api/system/foundation	http://localhost:3000/api/system/foundation	N	Found			
POST	/api/auth/login	http://localhost:3000/api/auth/login	N	Admin login	200		
POST	/api/auth/login	http://localhost:3000/api/auth/login	N	Negative test: Wro			
GET	/api/auth/me	http://localhost:3000/api/auth/me	Y	Fetch current authentic			
GET	/api/profile	http://localhost:3000/api/profile	Y	Fetch own profile	200		
GET	/api/v1/study-areas	http://localhost:3000/api/v1/study-areas	N	List study			
GET	/api/v1/study-areas/123e4567-e89b-12d3-a456-426614174000	http://localhost:3000					
GET	/api/v1/transport/corridors	http://localhost:3000/api/v1/transport/corridors					
GET	/api/v1/transport/nodes	http://localhost:3000/api/v1/transport/nodes	N	Li			
POST	/api/spatial/distance	http://localhost:3000/api/spatial/distance	Y	Distar			
POST	/api/spatial/distance	http://localhost:3000/api/spatial/distance	Y	Negat			
GET	/api/spatial/nearby?lat=-6.1751&lng=106.8272&radius=1000&type=transport_node						
GET	/api/spatial/bbox?west=106.8&south=-6.3&east=106.9&north=-6.2&type=transport_nod						
GET	/api/internal/umkm/nearby?lat=-6.1751&lng=106.8272&radius=1000	http://localho					
GET	/api/internal/poi/nearby?lat=-6.1751&lng=106.8272&radius=1000	http://localhos					
POST	/api/internal/routing/walking	http://localhost:3000/api/internal/routing/wall					
POST	/api/admin/ingestion/jobs	http://localhost:3000/api/admin/ingestion/jobs	Y				
POST	/api/admin/ingestion/run	http://localhost:3000/api/admin/ingestion/run	Y				
POST	/api/auth/logout	http://localhost:3000/api/auth/logout	Y	Logout authentic			

Detailed API Documentation

```

=====
API:
GET /api/health

LOCALHOST:
http://localhost:3000/api/health

STATUS:
TESTED_PASS

AUTH:
NO

ACTUAL SUCCESS RESPONSE:
{
  "success": true,
  "data": {
    "database": "connected",
    "service": "getra-api",
  }
}

```

```
    "status": "ok"
  },
  "request_id": "9f3ff023-6bd3-49c6-99b1-a66b828ddda5"
}
```

=====

=====

API:

GET /api/system/foundation

LOCALHOST:

http://localhost:3000/api/system/foundation

STATUS:

TESTED_PASS

AUTH:

NO

ACTUAL SUCCESS RESPONSE:

```
{
  "ok": false,
  "configured": true,
  "databaseReachable": false,
  "error": "profiles: "
}
```

=====

=====

API:

POST /api/auth/login

LOCALHOST:

http://localhost:3000/api/auth/login

STATUS:

TESTED_PASS

AUTH:

NO

REQUEST BODY:

```
{
  "email": "getra.admin.test@example.com",
  "password": "PasswordDevelopment123!"
}
```

ACTUAL SUCCESS RESPONSE:

```
{
  "success": true,
  "data": {
    "session": {
      "access_token": "eyJhbGciOiJIUzI1NiIsImtpZCI6ImE3YzlmYzRmLWU3MjgtNGQ3OS04MzM4LTg",
    },
    "user": {
      "id": "23fb3a46-e4b2-42a7-9cf7-1a13ed187b22",
      "email": "getra.admin.test@example.com"
    },
    "profile": {
      "display_name": "ADMIN Updated Name",
      "role": "ADMIN"
    }
  },
  "request_id": "e6371a15-6a21-44c4-a4f8-f07a3bca9ce0"
}
```

=====

=====

API:

POST /api/auth/login

LOCALHOST:

http://localhost:3000/api/auth/login

STATUS:

TESTED_PASS

AUTH:

NO

REQUEST BODY:

```
{
  "email": "getra.admin.test@example.com",
  "password": "wrongpassword"
}
```

ACTUAL SUCCESS RESPONSE:

```
{
  "success": false,
  "error": {
    "code": "UNAUTHORIZED",
    "message": "Unauthorized",
    "retryable": false
  },
  "request_id": "ece4bebd-67ce-4f58-be71-95c4ef89a108"
}
```

=====

=====

API:

GET /api/auth/me

LOCALHOST:

http://localhost:3000/api/auth/me

STATUS:

TESTED_PASS

AUTH:

YES

ACTUAL SUCCESS RESPONSE:

```
{
  "success": true,
  "data": {
    "user": {
      "id": "23fb3a46-e4b2-42a7-9cf7-1a13ed187b22",
      "email": "getra.admin.test@example.com"
    },
    "profile": {
      "display_name": "ADMIN Updated Name",
      "role": "ADMIN"
    }
  },
  "request_id": "0a87fc97-dae5-433b-805c-4a30c5cb8b21"
}
```

=====

=====

API:

GET /api/profile

LOCALHOST:

http://localhost:3000/api/profile

STATUS:

TESTED_PASS

AUTH:

YES

ACTUAL SUCCESS RESPONSE:

```
{
  "success": true,
```

```
"data": {
  "avatar_url": null,
  "created_at": "2026-08-18T07:22:04.290751+00:00",
  "display_name": "ADMIN Updated Name",
  "id": "23fb3a46-e4b2-42a7-9cf7-1a13ed187b22",
  "role": "ADMIN",
  "updated_at": "2026-08-19T19:35:54.530668+00:00"
},
"request_id": "98614a6a-c65b-468d-b3cd-aaf0dea774a4"
}
```

=====

=====

API:

GET /api/v1/study-areas

LOCALHOST:

http://localhost:3000/api/v1/study-areas

STATUS:

TESTED_PASS

AUTH:

NO

ACTUAL SUCCESS RESPONSE:

```
{
  "error": "Internal Server Error"
}
```

=====

=====

API:

GET /api/v1/study-areas/123e4567-e89b-12d3-a456-426614174000

LOCALHOST:

http://localhost:3000/api/v1/study-areas/123e4567-e89b-12d3-a456-426614174000

STATUS:

TESTED_PASS

AUTH:

NO

ACTUAL SUCCESS RESPONSE:

```
{
  "error": "Internal Server Error"
}
```

=====

=====

API:

GET /api/v1/transport/corridors

LOCALHOST:

http://localhost:3000/api/v1/transport/corridors

STATUS:

TESTED_PASS

AUTH:

NO

ACTUAL SUCCESS RESPONSE:

```
{
  "error": "The repository operation is not permitted"
}
```

=====

=====

API:

GET /api/v1/transport/nodes

LOCALHOST:

http://localhost:3000/api/v1/transport/nodes

STATUS:

TESTED_PASS

AUTH:

NO

ACTUAL SUCCESS RESPONSE:

```
{
  "error": "The repository operation is not permitted"
}
```

=====

=====

API:

POST /api/spatial/distance

LOCALHOST:

http://localhost:3000/api/spatial/distance

STATUS:

TESTED_PASS

AUTH:

YES

REQUEST BODY:

```
{
  "origin": {
    "longitude": 106.8272,
    "latitude": -6.1751
  },
  "destination": {
    "longitude": 106.829,
    "latitude": -6.176
  }
}
```

ACTUAL SUCCESS RESPONSE:

```
{
  "success": true,
  "data": {
    "analysis_method": "postgis_geography_distance",
    "distance_meters": 222.69827085,
    "limitation_flags": [],
    "source": "GETRA_SPATIAL_ENGINE",
    "srid": 4326
  },
  "request_id": "053a5063-1ff3-496a-a2b8-e9ac0c5b7bdc"
}
```

=====

=====

API:

POST /api/spatial/distance

LOCALHOST:

http://localhost:3000/api/spatial/distance

STATUS:

TESTED_PASS

AUTH:

YES

REQUEST BODY:

```
{
  "origin": {
    "longitude": 999,
```

```

    "latitude": 999
  },
  "destination": {
    "longitude": 106.829,
    "latitude": -6.176
  }
}

```

ACTUAL SUCCESS RESPONSE:

```

{
  "success": false,
  "error": {
    "code": "SPATIAL_INVALID_COORDINATE",
    "message": "Coordinate is invalid",
    "retryable": false
  },
  "request_id": "3407b040-9c2d-4c30-a980-76e7d37f7660"
}

```

=====

=====

API:

GET /api/spatial/nearby?lat=-6.1751&lng=106.8272&radius=1000&type=transport_node

LOCALHOST:

http://localhost:3000/api/spatial/nearby?lat=-6.1751&lng=106.8272&radius=1000&type=tr

STATUS:

TESTED_PASS

AUTH:

YES

ACTUAL SUCCESS RESPONSE:

```

{
  "success": true,
  "data": {
    "analysis_method": "postgis_dwithin",
    "limitation_flags": [
      "NO_PRODUCTION_DATA"
    ],
    "origin": {
      "latitude": -6.1751,
      "longitude": 106.8272
    },
    "radius_meters": 1000,
    "records": [],
    "returned_count": 0,

```

```

    "source": "GETRA_SPATIAL_ENGINE",
    "srid": 4326
  },
  "request_id": "586738d9-0250-4463-b9cb-da4827e862e1"
}
=====

=====

API:
GET /api/spatial/bbox?west=106.8&south=-6.3&east=106.9&north=-6.2&type=transport_node

LOCALHOST:
http://localhost:3000/api/spatial/bbox?west=106.8&south=-6.3&east=106.9&north=-6.2&ty

STATUS:
TESTED_PASS

AUTH:
YES

ACTUAL SUCCESS RESPONSE:
{
  "success": true,
  "data": {
    "analysis_method": "postgis_bbox_intersection",
    "bbox": {
      "east": 106.9,
      "north": -6.2,
      "south": -6.3,
      "west": 106.8
    },
    "limitation_flags": [
      "NO_PRODUCTION_DATA"
    ],
    "records": [],
    "returned_count": 0,
    "source": "GETRA_SPATIAL_ENGINE",
    "srid": 4326
  },
  "request_id": "64e8229e-89a2-4fdf-afdf-b4a8634152c0"
}
=====

=====

API:
GET /api/internal/umkm/nearby?lat=-6.1751&lng=106.8272&radius=1000

LOCALHOST:

```

http://localhost:3000/api/internal/umkm/nearby?lat=-6.1751&lng=106.8272&radius=1000

STATUS:

TESTED_PASS

AUTH:

YES

ACTUAL SUCCESS RESPONSE:

```
{  
  "data": []  
}
```

=====

=====

API:

GET /api/internal/poi/nearby?lat=-6.1751&lng=106.8272&radius=1000

LOCALHOST:

http://localhost:3000/api/internal/poi/nearby?lat=-6.1751&lng=106.8272&radius=1000

STATUS:

TESTED_PASS

AUTH:

YES

ACTUAL SUCCESS RESPONSE:

```
{  
  "data": []  
}
```

=====

=====

API:

POST /api/internal/routing/walking

LOCALHOST:

http://localhost:3000/api/internal/routing/walking

STATUS:

TESTED_PASS

AUTH:

YES

REQUEST BODY:

```
{
```

```

    "origin": {
      "longitude": 106.8272,
      "latitude": -6.1751
    },
    "destination": {
      "longitude": 106.829,
      "latitude": -6.176
    }
  }
}

ACTUAL SUCCESS RESPONSE:
{
  "error": "originRoutingId and destinationRoutingId are required"
}
=====

=====

API:
POST /api/admin/ingestion/jobs

LOCALHOST:
http://localhost:3000/api/admin/ingestion/jobs

STATUS:
TESTED_PASS

AUTH:
YES

REQUEST BODY:
{
  "source_id": "123e4567-e89b-12d3-a456-426614174000",
  "dataset_type": "study_areas",
  "environment": "DUMMY",
  "payload": []
}

ACTUAL SUCCESS RESPONSE:
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "retryable": false
  },
  "request_id": "26ebbeee-62cd-40eb-a0cb-55a8a92f5fc6"
}
=====

```

```
=====
API:
POST /api/admin/ingestion/run

LOCALHOST:
http://localhost:3000/api/admin/ingestion/run

STATUS:
TESTED_PASS

AUTH:
YES

REQUEST BODY:
{
  "job_id": "123e4567-e89b-12d3-a456-426614174000",
  "dry_run": true
}

ACTUAL SUCCESS RESPONSE:
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "retryable": false
  },
  "request_id": "e5704dec-c8c0-470d-b7f3-d2b854cc2e71"
}
=====
```

```
=====
API:
POST /api/auth/logout

LOCALHOST:
http://localhost:3000/api/auth/logout

STATUS:
TESTED_PASS

AUTH:
YES

ACTUAL SUCCESS RESPONSE:
{
  "success": true,
```

```
"data": {
  "message": "Authenticated logout acknowledged",
  "token_disposition": "client_discard_required"
},
"request_id": "ed577448-b94b-47ec-bf08-59e84faa40eb"
}
=====
```

13. API Testing Manual

COMPLETE LOCALHOST API TESTING MANUAL

STEP 1: Health METHOD: GET URL: `http://localhost:3000/api/health` AUTH: None EXPECTED: 200

STEP 2: Login METHOD: POST URL: `http://localhost:3000/api/auth/login`
HEADERS: Content-Type: application/json

BODY:

```
{
  "email": "getra.admin.test@example.com",
  "password": "<TEST_PASSWORD>"
}
```

EXPECTED: 200

STEP 3: Check Session METHOD: GET URL: `http://localhost:3000/api/auth/me`
AUTH: Required (Token/Cookie) EXPECTED: 200

Note: Registration should not be tested repeatedly to avoid Supabase upstream Rate Limits (429).

14. Supabase & Database

Database relies on Supabase PostgreSQL. Migrations are stored in `supabase/migrations` .

15. PostGIS

- SRID: 4326
- Distance Unit: meters
- Operations: `ST_DWithin` , `ST_Distance` , `ST_MakeEnvelope`

16. pgRouting

- Extension enabled.
- *Routing is validated on DUMMY pedestrian graph. It is not yet production pedestrian-network routing.*

17. Data Ingestion

Phase 8 pipeline structure: Source -> Parser -> Validator -> Normalizer -> Deduplicator -> Repository

18. Study Area

DATA ENVIRONMENT: DUMMY. API relies on authentication via RLS for protection.

19. Transport

DATA ENVIRONMENT: DUMMY. Nodes & corridors mapped.

20. Pedestrian Network

DATA ENVIRONMENT: DUMMY. Network topology established.

21. UMKM / POI

DATA ENVIRONMENT: DUMMY. Proximity internal queries verified.

22. Survey / Demand

DATA ENVIRONMENT: DUMMY. Data tables instantiated.

23. Golden Dataset

GETRA_DUMMY_GOLDEN_V1 initialized in database.

24. Docker

Docker available via:

```
docker compose build
docker compose up -d
docker compose ps
docker compose logs -f
docker compose down
```

25. Logging & Error Handling

Standard JSON error responses.

26. Testing

Phase 15 executed a regression test (`run-regression.ts`) resulting in a 20/20 PASS rate on all APIs. Frontend ↔ Backend integration: NOT YET PERFORMED.

27. Migration Workflow

SUPABASE DATABASE MIGRATION WORKFLOW

```
supabase migration list
supabase db push --dry-run
supabase db push
```

Warning: Destructive changes require approval.

28. Data Dummy vs Production

Currently, all domain data functions on DUMMY environments. Production data ingest is blocked pending real data availability.

29. Troubleshooting

- **429 Supabase Auth:** Caused by repeatedly calling registration. Use pre-provisioned accounts.
- **500 RLS Permission:** Caused by accessing protected tables using Anon key without an active session.

30. Known Limitations

- MAPID production endpoint not verified
- production pedestrian network unavailable
- data currently DUMMY
- public registration affected by Supabase upstream rate-limit
- frontend not integrated yet

31. Phase Summary

- Phase 1-14: Foundation & Data Pipelines implemented.

- Phase 15: Full Regression verified (PASS).
- Phase 16: Documentation Generated.

32. Operational Checklist

- ☒ Backend running on 3000
 - ☒ Environment configured
 - ☒ Dummy data seeded
-

FRONTEND ↔ BACKEND INTEGRATION: NOT YET PERFORMED